

Introduction to

Ubuntu Linux OS

An Academic Introduction to the World's Most Popular Open-Source Operating System

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What is an Operating System?

Definition

- An Operating System (OS) is system software that manages computer hardware and software resources, providing common services for application programs.

Core Functions

- Process Management — schedules and runs programs concurrently
- Memory Management — allocates and protects RAM for processes
- File System Management — organises data on storage devices
- Device Management — communicates with hardware via drivers
- Security & Access Control — enforces user permissions

Examples of Operating Systems

- Windows, macOS, Linux distributions, Android, iOS

History & Origins of Linux

1969

- UNIX created at Bell Labs by Ken Thompson & Dennis Ritchie

1984

- GNU Project launched by Richard Stallman – free software movement

1991

- Linus Torvalds releases Linux kernel 0.01 from Helsinki, Finland

1994

- Linux kernel 1.0 released – stable, production-ready

2004

- Ubuntu 4.10 'Warty Warthog' – first Ubuntu release by Canonical

2025

- Ubuntu remains the #1 Linux distribution globally

What is Ubuntu?

Overview

- Ubuntu is a free, open-source Linux distribution developed and maintained by Canonical Ltd., founded by South African entrepreneur Mark Shuttleworth in 2004.

The Name

- "Ubuntu" is an African philosophy meaning "I am because we are" — reflecting community-driven development and collaboration.

Release Cycle

- New version every 6 months: April (YY.04) and October (YY.10)
- Long-Term Support (LTS) releases every 2 years with 5 years of official support

Backing

- Canonical Ltd. provides commercial support, cloud services, and enterprise solutions built on Ubuntu.

Ubuntu Editions & Variants

Ubuntu Desktop

Standard workstation edition with GNOME DE. Ideal for personal computers and developers.

Ubuntu Server

Minimal server edition without a GUI. Used for web, database, and cloud deployments.

Ubuntu Core

Immutable, container-friendly OS for IoT devices and embedded systems.

Kubuntu

Ubuntu with KDE Plasma desktop — highly customisable, Windows-like interface.

Xubuntu

Ubuntu with XFCE desktop — lightweight, ideal for older or low-spec hardware.

Ubuntu LTS

Long-Term Support (e.g. 24.04 LTS). 5 years support; recommended for production.

Ubuntu Architecture

▲ User
Space ▲

Applications & GUI

Desktop Environment (GNOME)

System Libraries (glibc, GTK)

Shell & System Daemons (systemd, bash)

Linux Kernel (process, memory, I/O)

Hardware (CPU · RAM · Storage · Network)

Key Features of Ubuntu

Security & Stability

- Built-in firewall (ufw), AppArmor mandatory access control, and regular security patches. No antivirus required for most use cases.

Free & Open Source

- Ubuntu is free to download, use, and modify. Source code is publicly available under GPL and other open-source licences.

Vast Software Repository

- Over 50,000 packages available via APT (Advanced Package Tool). One-command installation of virtually any software.

Multi-User & Networking

- Designed as a multi-user OS with strong network support — SSH, Samba, NFS, and cloud integration built in.

Hardware Support

- Supports x86, ARM, RISC-V, and more. Excellent driver compatibility with modern and legacy hardware.

The Ubuntu Desktop Environment (GNOME)

GNOME Shell

- Ubuntu 17.10+ uses GNOME as its default DE. Activities Overview, a dock, and workspace management provide a modern, unified workflow.

Key UI Components

- Top Bar — clock, notifications, quick settings, battery & network
- Dash to Dock — pinned and running application launcher
- Activities Overview — window switcher, workspace grid, app search
- GNOME Files (Nautilus) — graphical file manager

Customisation

- Themes, icon packs, and extensions available via GNOME Extensions and the Ubuntu Software Centre, enabling personalised workflows.

Accessibility

- Screen reader (Orca), zoom, on-screen keyboard, and high-contrast themes are built in.

The Terminal & Command Line Interface

Terminal – bash

```
user@ubuntu:~$ ls -la
user@ubuntu:~$ sudo apt update
user@ubuntu:~$ pwd
user@ubuntu:~$ mkdir my_project
user@ubuntu:~$ nano README.txt
user@ubuntu:~$ _
```

Essential Commands

ls

List directory contents

cd

Change directory

pwd

Print working directory

cp / mv / rm

Copy, move, delete files

sudo

Run with superuser privileges

man

Display command manual page

grep

Search text using patterns

Package Management & Software Installation

APT — Advanced Package Tool

- Ubuntu uses APT to install, update, and remove software. Packages are sourced from Canonical's official repositories and third-party PPAs (Personal Package Archives).

Common APT Commands

- `sudo apt update` — refresh package lists
- `sudo apt upgrade` — install all available upgrades
- `sudo apt install <pkg>` — install a specific package
- `sudo apt remove <pkg>` — remove a package

Snap Packages

- Canonical's Snap system provides containerised, self-updating app bundles that work across any Linux distribution.

Ubuntu Software Centre

- A graphical front-end to APT and Snap, allowing users to browse and install applications without the terminal.

Ubuntu in Practice: Uses & Applications

Web Servers

Apache & Nginx run on Ubuntu. Ubuntu Server powers ~35% of public-facing web servers worldwide.

Cloud Computing

AWS, Google Cloud, and Azure all offer Ubuntu as a first-class VM image. Canonical certifies Ubuntu for cloud use.

Data Science & AI

Python, TensorFlow, PyTorch, Jupyter, and CUDA all have first-class support on Ubuntu.

Education

Ubuntu is widely used in university labs, Raspberry Pi projects, and coding bootcamps for its free, open nature.

Embedded & IoT

Ubuntu Core powers smart devices, robots (ROS), and industrial IoT appliances.

Developer Workstations

Git, Docker, VSCode, and development toolchains install natively. Preferred OS at many tech companies.

Advantages & Limitations

✓ Advantages

- Free of charge with no licensing fees
- High security — less malware exposure
- Large, active community & documentation
- Excellent performance on modest hardware
- Fully customisable and open source
- Strong privacy — no telemetry by default
- Ideal for programming and server use

✗ Limitations

- Steeper learning curve for new users
- Some proprietary software unavailable
- Limited gaming support vs. Windows
- Hardware compatibility gaps (some Wi-Fi/GPU)
- Enterprise support requires paid Canonical contract
- Snap app performance can lag vs. native
- Inconsistent UI between app ecosystems

Summary & Conclusion

What We Covered

- The role of operating systems and how Ubuntu fits within the Linux ecosystem
- Ubuntu's history from UNIX and the GNU/Linux kernel to Canonical's 2004 launch
- Architecture, desktop environments, and command-line interface
- Package management with APT and Snap
- Real-world applications across servers, cloud, AI, education, and IoT

Key Takeaway

- Ubuntu offers a powerful, free, and highly versatile platform suitable for beginners and professionals alike. Its open-source nature, strong community, and enterprise backing make it the ideal starting point for learning Linux.

Next Steps

- Try Ubuntu: download from ubuntu.com and run it live or in VirtualBox

Thank You

Questions & Discussion

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